Fieldnotes

HyLogger Data Bridge

A conversation with a durable record of the investigation

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GSWA HyLogger Hackathon 2026 · Data Adoption

The work between data and a decision

“What changed in this hole?”

One geological question crosses several kinds of work.

Data understanding

What is present, missing or excluded?

Scientific computation

Which comparison can the data support?

Expert review

What deserves another look, and why?

Investigation history

Can we recover the evidence and reasoning?

The investigation lifecycle

Chat agent

Coordinates the investigation

Python toolkit

Calculates reproducible evidence

Geologist

Chooses the next direction

01

Question

Natural language

02

Data and QA

pytsg and calculations

03

Evidence

Live browser book

04

Expert review

Choice and exact note

05

Next step

Agent continuation

Local Qwen advice Suggestions stay separate from evidence and expert choices.

A real case: SWMB007s

Are these intervals suitable to compare, and what differs?

Interval	Valid spots	Coverage	Primary TSG group
0–5 m	5/5	5/5	Kaolin: 5/5
5–10 m	5/5	5/5	Smectite: 5/5
10–15 m	5/5	5/5	Smectite: 5/5

First comparison: different labels

Changed intervals: the same labels

TSG interpretations only. Five spots per interval do not establish a geological boundary or grade.

Source: WA NVCL, NCI THREDDS · SWMB007s · DOI 10.25914/ctaf-6c13 · CC BY 4.0.

The expert directs the next page

Continue

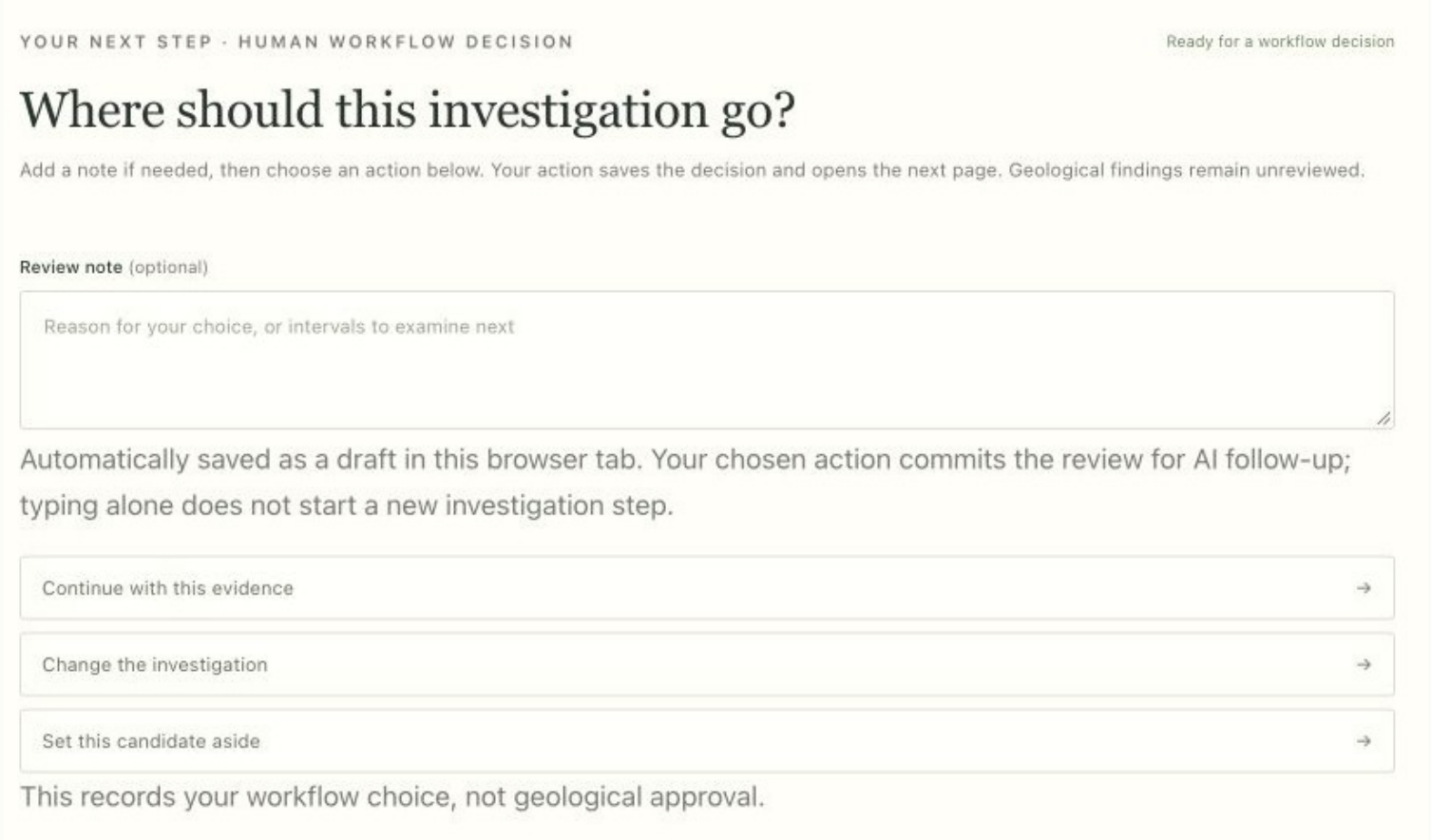
Examine the initial comparison.

Change

Compare 10–15 m against 5–10 m.

Set aside

Retain the evidence and stop this follow-up.



Each committed choice preserves the evidence and exact note.

What the technical pilot showed

Three independent agents. One real hole. The same six requests.

Condition	Numerical checks	Review record
Chat + Python/pytsg	25/25	Custom JSON records
Chat + toolkit	25/25 after retry	4 pages, replay passed
Chat + toolkit + Qwen	25/25	4 pages, replay passed

Qwen advice showed no added benefit

All three diagnostic suggestions were “clarify”. A no-advice startup defect required retry. This pilot did not measure expert time savings or geology accuracy.

Methods and full results: developer/docs/evaluation.md in the project repository

A usable investigation record

Keep the conversation.
Keep the evidence behind each next step.

Available preview

Apple Silicon installer
Codex tools and live evidence book
Real HyLogger data through pytsg

Next validation

Expert usefulness in practice
Broader data and stronger advice
Additional chat hosts

github.com/david3xu/hylogger-data-bridge